Pham Hung Quoc Viet

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OBJECTIVE

Proactive 3rd-year student at University of Information Technology – VNU-HCM (UIT) with practical experience in Big Data Processing, Data Warehousing, and Machine Learning. Adept at building robust data pipelines and translating complex datasets into actionable strategic insights. Seeking an internship position as a **Data Analyst, Data Engineer, or Business Analyst** to leverage technical proficiency and drive impactful, data-driven business solutions.

EDUCATION

University of Information Technology – VNU-HCM (UIT) — 2023 – 2027 (expected)
Bachelor of Information Systems — GPA: 8.06/10

SKILLS

Programming	Python, Java, C++, PHP, SQL, JavaScript, HTML5, CSS
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn, Plotly, Statistical Analysis, EDA, Data Cleaning & Wrangling, Excel
Databases	SQL Server, PostgreSQL, MySQL, Oracle
Big Data & BI	Apache Spark (PySpark), Kafka, SSIS, SSAS, Power BI, MDX
AI & Machine Learning	Scikit-learn, TensorFlow, XGBoost, Feature Engineering, Ensemble Learning, Anomaly Detection, SMOTE, SHAP, Data Mining
Tools & Platforms	Docker, Git, GitHub, SAP S/4HANA, Odoo, Kaggle, Streamlit, draw.io, UML, Figma
Others	Agile/Scrum Methodology, IELTS 6.5 (2022)

PROJECTS

NYC Taxi — Real-time Fleet Intelligence & Anomaly Detection — 2026

PySpark, Kafka, Docker, PostgreSQL, Streamlit, Pandas, Scikit-learn | GitHub

- **Data Ingestion:** Configured Kafka & Zookeeper clusters; wrote Python producers to push JSON events at 200 msg/s to a centralised topic.
- **Stream Processing:** Used PySpark Structured Streaming (5,000 offsets/trigger) to engineer 14 real-time features using Spark SQL and Pandas.
- **Anomaly Detection:** Built Isolation Forest + MinCovDet ensemble model with strict-voting to minimise false positives.
- **Monitoring & Retraining:** Output results to PostgreSQL; built a Streamlit dashboard with automated model retraining every 25 batches.

E-commerce Consumer Behaviour Analysis & BI Solution — 2025

SSIS, SSAS, SQL Server, Power BI, MDX, Python, Scikit-learn | GitHub

- **ETL Pipeline:** Developed automated SSIS packages to extract, clean, and migrate raw CSV datasets into SQL Server Data Warehouse.
- **OLAP Modelling:** Designed Star Schema with Fact and 5 dimensions; built SSAS Cubes with MDX for Drill-down and Roll-up analysis.
- **Data Mining:** Implemented K-Means clustering for customer segmentation and Decision Trees to predict spending levels using Python (Scikit-learn).
- **Visualisation:** Created Power BI dashboards integrating OLAP cubes and ML predictions to monitor customer retention KPIs.

Bank Customer Churn Prediction & Explainability — 2026

Python, Scikit-learn, TensorFlow, XGBoost, LightGBM, SMOTE, SHAP, Optuna, Pandas, Seaborn | GitHub

- **EDA & Feature Engineering:** Conducted EDA on a 10k-record European bank dataset; applied Winsorisation, encoding, normalisation, and engineered new features to improve model signal.
- **Class Imbalance Handling:** Applied SMOTE to address 80/20 class imbalance (churn rate 20.4%), augmenting training data from 6,400 to 10,192 samples and boosting recall.
- **Multi-Model Benchmarking:** Trained and compared 7+ algorithms (XGBoost, Random Forest, LightGBM, etc.) with GridSearchCV and Optuna tuning.
- **Explainability (XAI):** Applied SHAP and LIME; Jaccard similarity analysis (0.596) identified Age, IsActiveMember, and NumOfProducts as key churn drivers.